

ESIG ADVANCED GRID SOLUTIONS WORKSHOP

INTEGRATING GETS INTO TRANSMISSION PLANNING AND OPERATIONS : A UTILITY FRAMEWORK

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Xcel Energy System Footprint

Xcel Energy serves eight states — the operational and planning context for this framework

- Electric service across **eight states** with diverse generation, load, and interconnection profiles
- Approx 4 million electric and 2.5 million natural gas customers. Opcos across MISO, SPP and Non-RTO

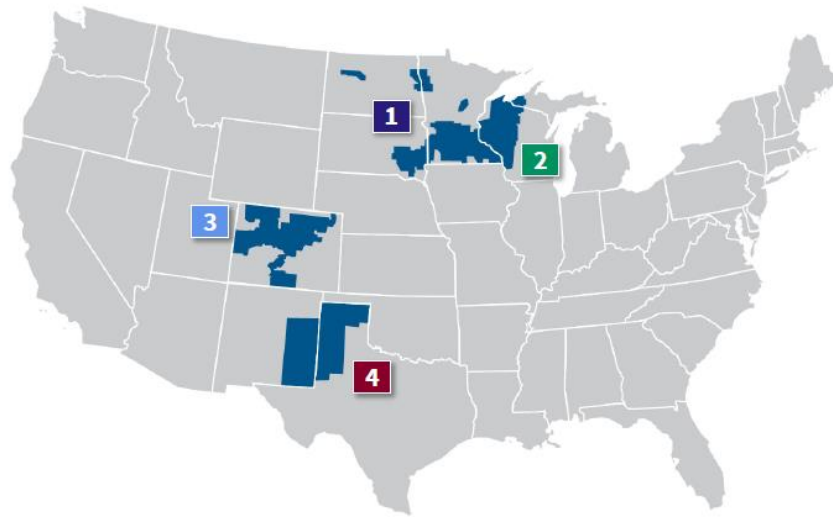
Planning context: Xcel Energy's transmission system underpins all GETs screening, case studies, and governance frameworks discussed in this session.

1 Northern States Power Company - Minnesota (NSPM)

2 Northern States Power Company - Wisconsin (NSPW)

3 Public Service Company of Colorado (PSCo)

4 Southwestern Public Service (SPS)



FERC REGULATORY DRIVERS FOR GETS

Three FERC orders collectively mandate that GETs be evaluated as first-class options in planning, interconnection, and real-time operations



Order No. 1920

Long-Term Regional Transmission Planning

DLR **APFC** **TTO** **High Perf. Conductors**

- Requires transmission providers to consider GETs in long-term regional transmission planning
- Specifically names DLR, APFC devices, Transmission Switching/Topology Optimization, and High Performance Conductors
- 1920 requires to evaluate if GETs can solve the same Needs more efficiently or cost-effectively than other potential facilities. Order 1920 is a little more complex to determine if GETs can qualify for regional cost allocation. Minimum criteria includes $\geq 200\text{kV}$



Order No. 881

Ambient-Adjusted Ratings for Planning & Dispatch

Hourly Temp-Adjusted Ratings **15-min DLR**

- Requires utilities to use hourly temperature-adjusted line ratings for planning and dispatch
- RTO/ISOs must accept 15-minute ratings from transmission owners
- Eliminates purely seasonal static ratings — weather-responsive ratings are now the standard



Order No. 2023

Generator Interconnection Process Reform

APFC **High Perf. Conductors** **DLR (eligible)**

- GI processes must evaluate APFC devices and high performance conductors
- DLR and topology optimization are optional but eligible for credit in constrained corridor studies
- Reshapes how generator interconnection studies handle capacity-constrained corridors

The Transmission Problem — Timing Mismatch and GETS Role

Conventional builds remain essential, but multi-year lead times create a near-term gap that GETs might be positioned to fill under certain circumstances

- Load growth, IBR interconnection, and resource mix shifts are **accelerating simultaneously** across planning horizons
- Conventional construction requires routing, permitting, siting, procurement, and build — **multi-year timelines are structural**, not exceptional
- GETs **might be able to** unlock **latent capacity in existing corridors**, bridging near- and mid-term reliability and congestion needs
- GETs must be screened **alongside not instead of** traditional upgrades



Near-term capacity gap while conventional build progresses

GETs deliver operational relief during this window — not as a permanent substitute

Four planning needs GETs address:

Reliability

Congestion

Interconnection

Affordability



GETs Portfolio — Four Families and Their Engineering Roles

Each GET family solves a distinct engineering problem; matching constraint type to GET family is the first screening discipline

⚠ Screening Rule: Constraint pattern determines first-fit GET — not device availability

1



Advanced Conductors

ACCC and equivalents

Increase ampacity on constrained corridors while managing sag, clearance margins, and structure load limits.

Structural Capacity

2



Dynamic Line Ratings

DLR / AAR

Reflect changing ambient conditions through adjusted, capped, or real-time ratings with defined fallback limits. Uses IEEE / CIGRE thermal balance models.

Risk-Based Utilization

3



Advanced Power Flow Control

APFC

Redirect MW flow from overloaded paths to parallel paths with available headroom via controllable impedance or phase-angle injection.

Controllability

4



Topology Optimization

TTO / Transmission Switching

Evaluate repeatable switching configurations that reduce congestion while maintaining N-1 reliability.

Operational Flexibility

Advanced Conductors — Engineering Deep Dive and Xcel Case

The \$87M capital savings case demonstrates that advanced reconductoring can deliver 60%+ ampacity gain at ~25% of new-line cost, but utility validation — not vendor data — determines what can be counted.

\$80-90M

Capital Savings
(Xcel Case)

+50-60%

Ampacity Increase
936 A → 1,520 A

30 mo.

Schedule Reduction
48 mo. → 18 mo.

Xcel Case — Cost Comparison

Conventional Rebuild

\$135M

Reconductoring (adv. cond)

\$48M

✓ ~25% of conventional new-line cost

⚠ Maturity evidence supports alternatives screening — utility studies determine what rating can be counted in reliability and economic models

Vendor-Reported Benchmarks (Require Utility Validation)

Performance Metric	Vendor Range	Utility Action Required
Operating capacity increase	50–100%	Validate with site-specific thermal model
Thermal sag reduction	~50%	Verify clearance margins at critical spans
Conductor resistance reduction	~30%	Update R, X params in PSSE/PSLF/TARA
Line loss reduction	Up to 40%	Confirm in PROMOD production cost model
Indicative process duration	8–24 months	Validate against actual permitting schedule

All ranges are vendor-reported — utility studies determine defensible planning values

Technology Maturity — Global Deployment Evidence

125K+

Conductor miles installed worldwide

1,450+

Completed projects (100+ active)

300+

Utilities across 65+ countries

241

Reported N. America projects

Planning Role & Best-Fit Indicators

Deploy where sustained thermal or clearance constraint exists and corridor expansion is infeasible. Best fit: constrained ROW, clearance-limited spans, GI-driven incremental capacity needs.

Modeling Requirements

Update thermal ratings and conductor electrical parameters in **PSSE PSLF TARA PROMOD** — verify voltage profile and N-1 contingency performance.

Key Risks

Terminal equipment ampacity limits · Structure mechanical life · Installation quality · Ice/galloping and vibration response

Advanced Conductors — Xcel Deployment Experience

With ~tens of miles installed since 2005 and 11 new projects approved, Adv. cond is Xcel's most institutionalized GET — but structural validation of aging wooden poles and wildlife ROW constraints define the deployment envelope.

Xcel ACCC Deployment Timeline

- 2005**
First Advanced Conductor installation at Xcel
- Current**
Many miles of advanced conductors installed across PSCo and broader Xcel system
- Pipeline**
Many approved advanced reconductoring projects in the pipeline

~tens of miles
Installed ACCC

Multiple
New Projects Approved

20+ yr
Operational History

ACCC Applicability
Favorable

- ✓ Constrained ROW
- ✓ Sensitive wildlife areas
- ✓ GI-driven incremental need
- ✓ Clearance-limited spans

Unfavorable

- ✗ End-of-life wooden poles without remediation
- ✗ Substation equipment as binding constraint

Primary Deployment Driver

- **GI-driven incremental capacity needs** — marginal additional headroom required without new ROW acquisition; wildlife-protected areas where corridor expansion is physically or permitting-constrained make ACCC the preferred option
- **Structural Validation Requirement**
Engineering designers must validate that existing structures have sufficient capacity and remaining lifespan before reconductoring. Old wooden poles near end-of-life are **not structurally appropriate** for Adv. cond without remediation — this is a non-negotiable prerequisite, not a planning assumption.
- **Design Standards Opportunity**
Opportunity within Xcel Energy to integrate Adv. cond into construction design standards for engineering contractors — enables cost-benefit consideration at the design stage, not only during planning studies. Adv. cond is the simplest GET to incorporate due to seamless integration with existing planning tools and operational expectations.
- **Planning tool compatibility:** Adv. cond integrates seamlessly with PSSE, PSLF, TARA, and PROMOD (basically all software tools) — update thermal ratings and conductor electrical parameters (R, X); verify voltage profile and N-1 contingency performance.

Advanced Conductors — Cost Economics & Industry Benchmarks

Adv. cond's 2.5–5x material cost premium over ACSR is justified only when lifecycle economics — including avoided rebuild cost, schedule savings, and long-term thermal aging — are fully modeled.

Adv. cond Material Cost Premium vs. ACSR

2.5–5x

Warrants justification through full lifecycle economics — not vendor assertion

Premium offset by: avoided structure replacement, ROW acquisition, and construction complexity. Utility validation required.

Global Industry Scale — Technology Maturity

125,000+

Installed conductor miles worldwide

1,450+

Completed projects (100+ active)

300+

Utilities across 65+ countries

241

Reported North America projects

Lifecycle Considerations — Risk & Engineering Controls

Risk Category	Engineering Control
Long-term thermal aging at elevated operating temperature	Conductor-specific thermal model; periodic inspection intervals per design standard
Vibration and galloping under ice/wind loading	Specialized vibration damper specification; fatigue analysis for critical spans
Installation quality — significant fraction of historical failures are installation-related	Contractor training and qualification; rigorous procurement specification; construction QA/QC protocol



Decision Discipline

Material cost premium is not the gating criterion — lifecycle economics over the planning horizon determines cost-effectiveness relative to conventional rebuild. Advanced conductors are primarily warranted for **constrained corridor reconductoring**, not new lines without space constraints.



DLR Planning Methodology — Making Ratings Bankable in Studies

A DLR rating only enters planning studies as a defensible assumption when five validation steps are completed and risk thresholds explicitly documented

- 1

Need
Confirm recurring thermal overload or binding congestion on candidate line
- 2

Physics
Apply IEEE Std 738 / CIGRE TB 601 thermal balance model; identify critical spans; verify clearance and terminal equipment limits
- 3

Data
Assemble historical weather record, local sensor data, forecast uncertainty quantification, bad-data flagging protocol
- 4

Planning Cap
Establish seasonal, monthly, or probabilistic rating by availability threshold (e.g., 90th-percentile ambient; P90 availability floor) — this number enters studies, not the instantaneous maximum
- 5

Operations
Confirm EMS integration path; define fallback ratings (revert to static seasonal); produce operator playbooks for rating transitions and alarm conditions

Planning Study Integration

 **Reliability planning:** Enter capped DLR limits in N-1 contingency studies (PSSE / PSLF / TARA / etc.)

 **Economic planning:** Enter conservative DLR rating profiles (seasonal/monthly caps) in production cost models (PROMOD / PLEXOS/ etc.)

Governance Gate — DLR Decision

Advance

Advance as Bridge

Hold — Refine

Escalate to Wires

Risk Controls to Document

 Rating accuracy band

 Sensor availability %

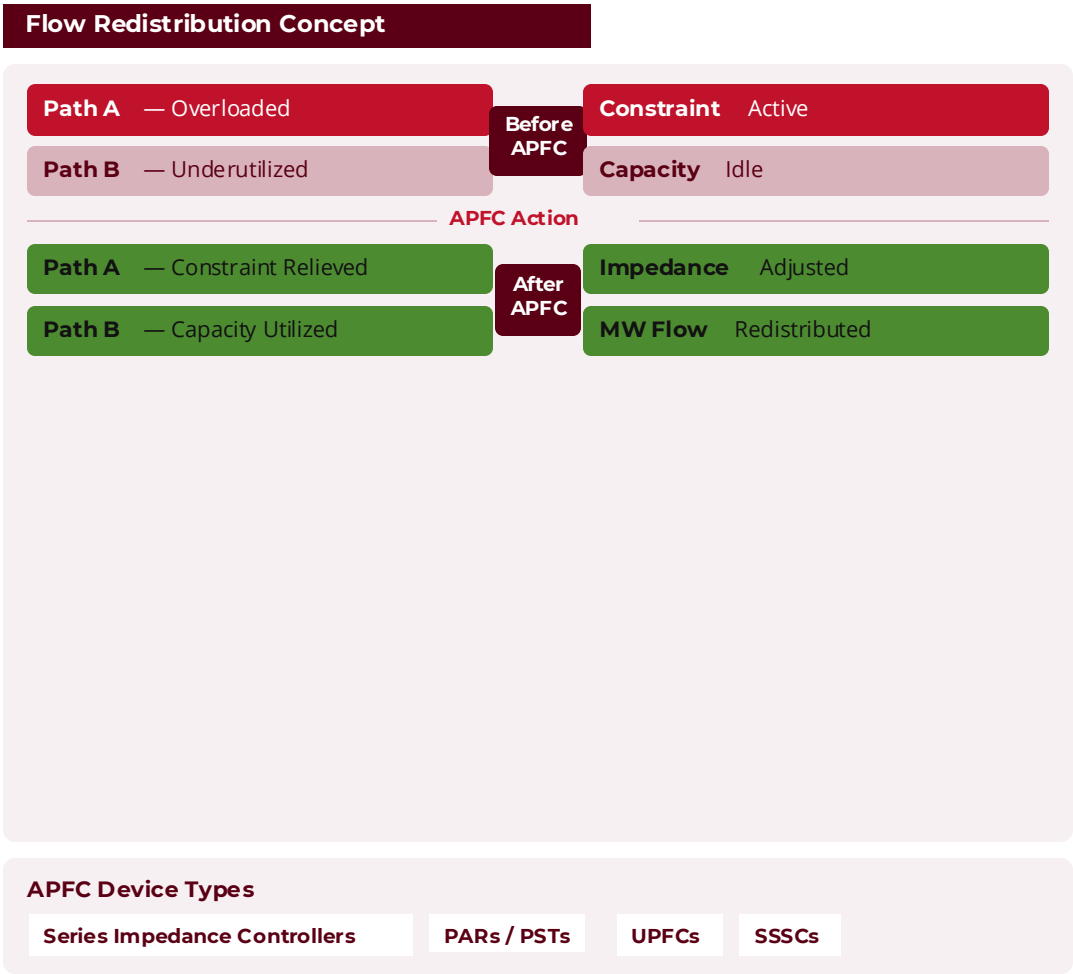
 Data quality flags

 Operator usability score

 Lifecycle maintenance plan

APFC Deep Dive — Flow Control Architecture and Study Requirements

APFC provides the only controllable MW redistribution among GET families, but value depends entirely on completing power-flow sensitivity, contingency, and dynamic interaction studies before deployment



 **Planning Role**

Right-size, phase, and potentially relocate APFC to match constraint evolution — avoids overcommitting to fixed infrastructure. Most deployments remain static or operator-directed; automatic closed-loop control is the exception.

 **Required Study Types**

- Power-flow sensitivity analysis (MW shift factors, PTDF impacts)
- N-1 and N-1-1 contingency performance
- Voltage impact assessment across affected buses
- Protection coordination review (distance relay reach, fault current changes)
- Dynamic interaction study

 **Practical Barriers to Deployment**

 Substation space requirements

 Control architecture complexity

 Cybersecurity classification

 EMS integration for real-time setpoint visibility

APFC — Industry Deployment Landscape

Most APFC deployments operate in static mode as of Spring 2026 — the operator trust gap and substation space constraints are the primary barriers to realizing full autonomous control value.

Current Deployment State — Spring 2026

Most APFC Devices Operate in Static Position

- As of Spring 2026, most publicly shared APFC deployments operate devices in a **static position** — not with automatic real-time adjustments
- Devices **can** respond to real-time conditions, but most control room operators **do not yet trust autonomous state changes**
- Even static deployment provides some value: configuration can be changed later as grid topology or congestion patterns evolve

PSCo Study Finding

APFC **sufficiently reduced loading** on a circuit, but redirecting flows caused **overloads on other circuits** — the core system-level interaction risk that must be modeled before deployment

Possible Use Cases

Use Case	RTO / ISO / Utility Context
Loop flow management and unscheduled flow mitigation	Some ISO / RTO footprints
Parallel path loading enforcement and contractual flow path control	Meshed network corridors
Interim congestion relief during major transmission construction outages	Bridge solution during construction

 **Operator Trust Gap:** Space exists for larger industry consensus and collaboration with control room operators to identify risk mitigation measures needed to develop confidence in autonomous APFC control

APFC — Path Forward for Xcel

Xcel's lack of APFC deployment reflects a disciplined sequence — governance, EMS integration, and operator confidence must be built before autonomous control is appropriate.

Implementation Path

1 Constraint Identification

Identify binding congestion or thermal overloads on candidate lines requiring controllable flow redistribution

2 Alternatives Analysis

Formal comparison: APFC vs. PST vs. conventional upgrade, including power electronics lifecycle costs

3 Governance Gate

Capital planning and project approval through Xcel's standard transmission governance framework

4 Pilot Static Deployment

Start with static position at a well-understood constraint — build reliability evidence before progressing to operator-directed control

5 Expand to Operator-Directed Control

Modular design enables relocation as constraints migrate; autonomous mode contingent on sustained operator confidence

Governance Framework Requirements

- Formal alternatives analysis: APFC vs. PST vs. conventional upgrade at each candidate site
- Benefit-cost analysis including power electronics lifecycle costs (typically shorter than electromechanical devices)
- Capital planning and project approval aligned with Xcel's standard transmission governance gates

EMS Integration Requirements

- Real-time setpoint visibility for operators in the control room
- Alarm configuration and abnormal condition response procedures
- Fallback strategy to defined static position if communications or control fails

Cybersecurity Requirements

- APFC devices require formal cybersecurity classification — NERC CIP applicability assessment before any deployment
- Communications architecture designed with defense-in-depth principles

Building Operator Confidence

Coordinate with control room operators throughout design and commissioning— not only at go-live. Right-size modular device to current constraint; design for portability as constraints migrate.



Topology Optimization Framework — Five Integration Steps

TTO value scales only when real-time data replaces day-ahead approximations and proven configurations are institutionalized across planning, outage, and market operations

- 1 Integrate Real-Time Data**
Use RT load, renewables dispatch, and system conditions
- 2 Streamline and Automate Studies**
Automate screening for simple reconfigurations; fast-track post-contingency-only cases through expedited review
- 3 Expand Eligible Use Cases**
Include stability-driven constraints beyond thermal overloads; voltage-constrained topologies are valid candidates
- 4 Improve Transparency**
Share assumptions, constraints, and rejection reasons with operators and market participants; provide visibility into flows and thermal limits
- 5 Build Library of Proven Actions**
Catalog validated switching configurations; integrate library into Transmission Planning, Outage Planning, and Market Operations workflows

⚠ Modeling Gap

Operational models use **node-breaker** representation; planning models use **bus-branch**. A translation layer is required before TTO configurations move from operations to planning studies.

⚙ Operator Constraint

Switching frequency, authorization levels, stability limits, and market coordination rules all bound how many TTO actions can be executed in a given operating period.

📁 Integration Targets

Transmission Planning · Outage Planning · Market Operations

TTO — SPP/MISO Integration & Xcel's Path

SPP and MISO have proven that TTO delivers repeatable congestion savings (sometimes) / helps in outage coordination (sometimes)— Xcel is looking at solutions and the node-breaker to bus-branch translation gap define the implementation path for NSP, SPS, and PSCo.

SPP

Implemented — Active TTO

Implements TTO analysis and applies identified reconfigurations for outage coordination. **Publishes reconfiguration lists** with associated benefits for operator and planning use.

MISO

Implemented — Proven Case Study

161 kV corridor case (Sep–Dec 2024): **~\$24.5M savings** and **~80% congestion reduction** . Demonstrates repeatable, quantifiable TTO value.

NSP & SPS

Opportunity — Use RTO Outputs

If SPP and MISO publish identified reconfigurations alongside benefits, both NSP and SPS can implement those reconfigurations in planning models — evaluating as **alternatives or near-term bridges** to transmission projects.

⚠ Node-Breaker vs. Bus-Branch Model Gap

Node-Breaker
EMS / Operations

↔
Translation
Layer Required

Bus-Branch
Planning Models

Cannot plug-and-play: RTO-identified optimal reconfigurations may not be "optimal" in the planning model horizon — but can still be evaluated for planning-period performance.

Fundamental barrier: This gap is the primary technical obstacle to directly using RTO TTO outputs in Xcel planning studies.

🔌 PSCo EMS TTO Readiness

PSCo EMS does **not currently support TTO analysis**. Prerequisites for EMS TTO support must be confirmed before operational deployment.

⚡ New Tool Adoption

NEW TOOL

- Xcel is looking at adopting new software tools — evaluates trade-offs between GETs and traditional solutions in **steady-state reliability analyses**
- Allows consideration of TTO: **line switching status** (closing a normally open line or opening a line normally in service)
- Does **not yet support** full node-breaker TTO optimization — line switching status is the tractable scope

Modeling and Study Requirements Across All GET Families

Each GET family requires a distinct modeling representation and study sequence; skipping any step produces an indefensible rating or an unverifiable benefit claim

GET Family	Modeling Representation	Required Study Types	Primary Objective
Advanced Conductors	Updated thermal ratings; revised conductor electrical parameters (R, X); structural and clearance limit verification	Thermal rating, voltage profile, N-1 contingency (PSSE/PSLF/TARA)	Quantify structural ampacity gain and post-contingency performance
DLR / AAR	Adjusted static/seasonal/monthly caps and sensitivity cases (not instantaneous max)	Probabilistic thermal, risk-based reliability planning, production cost (PROMOD/etc.)	Quantify latent capacity and availability risk at defined confidence level
APFC	Controllable series elements modeled as impedance or phase-angle proxies in load-flow	Steady-state, N-1 contingency, voltage, dynamic stability and EMT as needed (PSSE/PSLF/PSCAD/ etc.)	Redirect flows without creating new overloads or protection conflicts
TTO	Breaker-status cases and alternate topology scenarios; node-breaker to bus-branch translation	Security-constrained power flow, SCOPF, economic analysis.	Identify repeatable switching actions with documented reliability margin

Tools Referenced:

PSSE

PSLF

TARA

PROMOD

Planning-to-Operations Continuum and Governance Gates

A GET delivers its rated benefit only when it is operationally visible, procedurally governed, and continuously monitored — the five-stage continuum and five decision gates enforce this discipline

5-Stage Planning-to-Operations Continuum

- 1 Forecast & Needs**
Load growth, resource queues, congestion history, IBR interconnection backlog
- 2 Alternatives Analysis**
Traditional upgrades plus GET portfolio screening (constraint-to-GET mapping)
- 3 Engineering Validation**
Thermal ratings, power-flow, stability, and protection coordination studies
- 4 Operations Readiness**
EMS integration, display configuration, operator procedures, and training
- 5 Performance Management**
Realized vs. planned benefits, model calibration, risk assumption updates, expansion triggers

5 Governance Decision Gates

- Gate 0 Need Confirmed**
Recurring overload or congestion above threshold documented
- Gate 1 Screened Fit**
GET family matched to constraint pattern; first-order feasibility met
- Gate 2 Validated Studies**
Ratings, contingency, and economic studies completed and peer-reviewed
- Gate 3 Operational Readiness**
EMS integration confirmed; fallback ratings defined; operator playbooks approved
- Gate 4 Scale or Escalate**
Performance data reviewed; decision to replicate, expand, or escalate to conventional upgrade

 Documentation standard: every advance or exclusion records technical basis, study assumptions, risk treatment, operating requirements, and escalation trigger

Evaluate total system value

Benefits to quantify

Congestion reduction

Curtailment reduction

Retirement deferral

Reliability margin

Operational flexibility

Restorations

Risks to govern

Implementation cost

Stranded risk

Maintenance burden

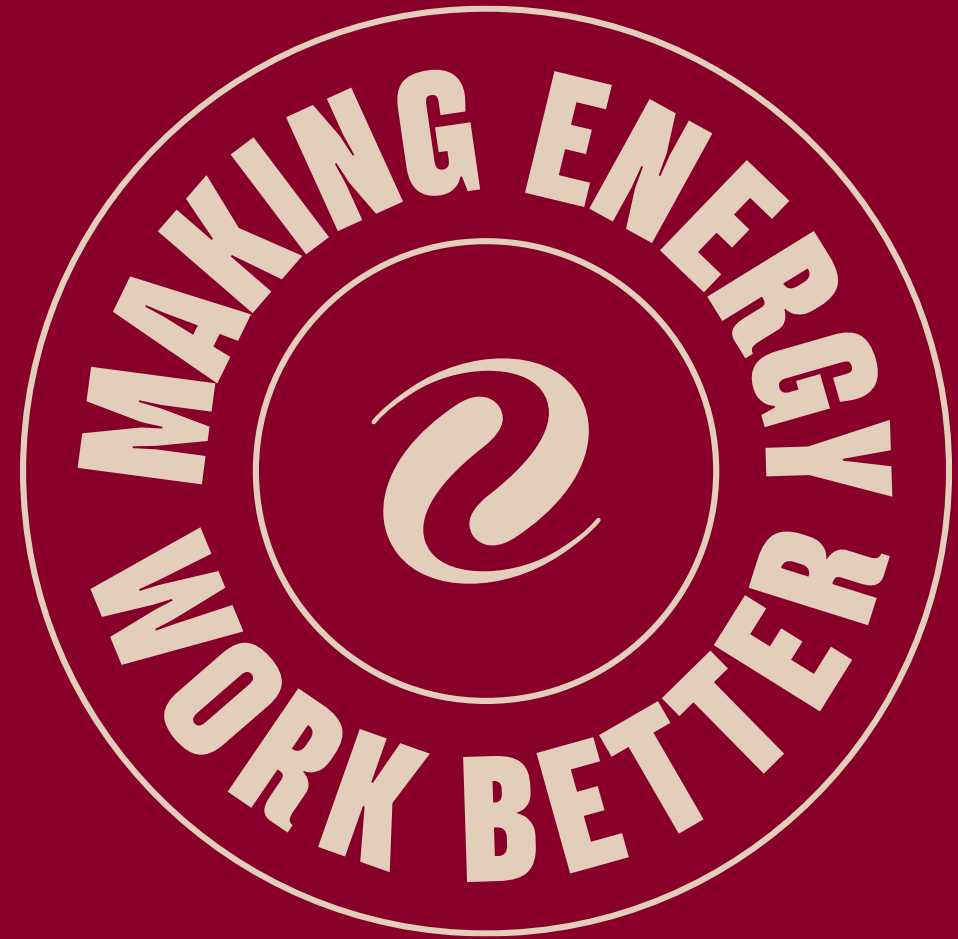
Cyber / data quality

Performance impacts

Operator trust

A GET should advance when the expected operating value exceeds lifecycle cost and risk, with clear accountability for assumptions.

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[xcelenergy.com](https://www.xcelenergy.com)

ADVANCED CONDUCTORS

Replacement of legacy conductors (e.g., ACSR) with high-temperature, low-sag (HTLS) advanced conductors.

Uses composite-core designs (e.g., ACCC, ACCR) enabling higher operating temperatures and lower sag.

Achieves higher ampacity without modifying ROW or structures in most cases.

Can often be implemented using standard installation practices and tools, minimizing operational disruption.

